

Requirements Table

REQ_ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE	COMMENTS
FR-001	Functional	The system shall allow diners to view the restaurant's live 2D floor plan and select an available table for a specified date and dining time.	high	Only available tables can be selected and reserved	Prevents double booking and allows diners to choose their preferred table	
FR-002	Functional	The system shall allow diners to browse the menu and add food items with quantities to their table reservation as a pre-order.	high	Selected menu items and quantities are associated with the reservation	Enables advance ordering and reduces waiting time after arrival	
FR-003	Functional	The system shall calculate the estimated kitchen preparation time for a diner's pre-order based on the selected food items and their preparation times.	high	The system displays an estimated preparation time before order confirmation	Helps coordinate food preparation with the diner's expected arrival	
FR-004	Functional	The system shall allow the restaurant manager to view and update the status of restaurant tables as Available, Reserved, Seated, or Billed.	high	The updated table status is displayed correctly on the manager's floor plan	Allows restaurant staff to maintain an accurate real-time view of table occupancy	
FR-005	Functional	The system shall allow the restaurant manager to view and manage advance food orders associated with reservations.	medium	The manager can view the reservation, ordered items, quantities, and preparation status	Allows the kitchen and restaurant staff to prepare orders before diners	

					arrive	
NFR-001	Performance	Table status changes shall synchronize across all manager floor terminals within 5s under the expected peak system load.	high	Performance testing demonstrates synchronization latency of ≤ 500 ms under simulated peak load	Fast synchronization is necessary to prevent multiple staff members from acting on outdated table information	
NFR-002	Security	The system shall require authenticated access for restaurant manager functions and shall prevent diners from accessing manager-only functions.	high	Unauthenticated users and diner accounts cannot access manager functions	Protects table management and advance-order information from unauthorized access	